SAFF-DETR: An End-to-End Object Detection Network for Remote Sensing Images With Targets of Varying Sizes Based on Scale Adaptation and Frequency Fusion

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***Abstract*—Deep learning-based object detection algorithms have achieved significant success in the field of computer vision. However, the wide range of target sizes in remote sensing images poses a challenge for single algorithms to detect objects of varying sizes effectively. To address this issue, this article proposes an end-to-end object detection algorithm for remote sensing images based on scale adaptive and frequency fusion DETR (SAFF-DETR), which designs a frequency feature enhancement and fusion mechanism to handle targets of varying sizes within a single framework. First, in order to improve the Transformer-based detectors' ability to detect small targets, a multibranch representation fusion (MRF) module is proposed to fuse shallow-layer frequency representations, boosting the network's ability to perceive small targets. Furthermore, cross-layer spatial–frequency attention (CSFA) and cross-layer channel–frequency attention (CCFA) are designed to enable efficient frequency feature interaction across multiscale features, enhancing the representation capability for targets of different sizes. Moreover, by integrating the two aforementioned attention mechanisms, the cross-layer channelwise–spatialwise frequency fusion (CCSFF) structure is introduced to realize global feature interactions in one step without repetitive upsampling and downsampling operations, by which a patch division-based Transformer architecture is designed to enhance scale adaptability for object detection. Experimental results over several benchmark datasets demonstrate that the proposed SAFF-DETR can handle extremely varying-sized targets and outperforms several state-of-the-art (SOTA) algorithms. The source code will be available at: https://github.com/Jianglin-Zhao/SAFF-DETR**

***Index Terms*—Frequency feature fusion, object detection, remote sensing image, Vision Transformer (ViT).**

I. Introduction

IN RECENT years, the rapid development of optical remote sensing technology has significantly improved the quality and quantity of remote sensing images [1]. Different imaging platforms (e.g., stallites and unmanned aerial vehicles (UAVs)) enable multiscale imaging capable of observations ranging from small regions to global coverage [2]. This makes optical remote sensing one of the most important methods for obtaining large-scale, high-resolution geospatial information, and it has been widely used in resource exploration, military operations, disaster prevention, urban planning, and ecological environment monitoring [3]. But, as shown in Fig. 1, due to factors such as imaging altitude and viewing angle, there can be significant variations in the sizes of objects of the same or different categories in remote sensing images. The large variation in object sizes makes it extremely challenging to simultaneously capture the global features of large objects and the fine details of small ones. This makes it difficult to accurately detect and recognize objects of different sizes within a single framework. Therefore, multisize detection has become a key issue that needs to be addressed in remote sensing object detection.

After years of technological development, remote sensing image object detection methods can be primarily categorized into two major types: traditional methods and deep learning-based methods. The former, often referred to as handcrafted feature-based methods, primarily relies on shape, texture information [4], salient features [5], and scale-invariant features [6] in images, combined with prior knowledge, to design suitable features for representing image content. While these methods can achieve remarkable results in specific scenarios, their performance is often limited when applied to unknown or dynamically changing environments. Particularly in cases where remote sensing images exhibit high diversity and complex variations, handcrafted features frequently fail to fully capture all critical information, significantly compromising the model's adaptability and generalization capabilities. The latter approach, on the other hand, leverages deep learning techniques, particularly convolutional neural networks (CNNs), which automatically extract relevant features through the hierarchical representation capabilities of deep network structures without requiring prior knowledge. These methods perform exceptionally well in classification and localization tasks [7], with prominent algorithms including two-stage detectors (e.g., Faster R-CNN [8] and Mask R-CNN [9]) and single-stage detectors (e.g., SSD [10] and the YOLO [11] family). How-

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Fig. 1. Objects in remote sensing images exhibit obvious size variations.

ever, these detectors typically depend on manually predefined anchor box sizes [12] and employ nonmaximum suppression (NMS) [13] for postprocessing. This not only increases the complexity of algorithm tuning but also hinders the model's ability to adaptively adjust to targets with wide-ranging size variations.

With the remarkable success of the Transformer [14] in natural language processing (NLP), researchers have begun to adapt it to the field of computer vision. Detection Transformer (DETR) [15] reformulates object detection as a set prediction problem, removing the need for region proposals and NMS postprocessing used in traditional detectors. Compared with traditional CNN-based methods, its streamlined architecture and end-to-end detection capability have garnered significant attention in the academic community, though this design results in reduced accuracy for small object detection and slow convergence. To tackle these challenges, subsequent researchers have proposed various improvements, such as architecture-based designs [16], [17], [18], [19], enhancements to object queries [20], [21], [22], training strategy optimizations [23], [24], [25], [26], [27], [28], [29], and loss function refinements [30], [31], significantly accelerating the development of DETR-like detectors. In 2023, Zhao et al. [32] introduced the real-time end-to-end Transformer Detector (RT-DETR). By optimizing the input encoder sequence and object queries, RT-DETR surpasses state-of-the-art (SOTA) YOLO detectors of comparable size in both speed and accuracy, demonstrating broad application potential. Although RT-DETR has achieved excellent results in the field of computer vision, when applied to remote sensing, it still faces typical problems represented by multisize object detection.

To deal with the challenges of significant target size variations, complex backgrounds, and imbalanced class distributions in remote sensing images, which lead to lower detection accuracy. Many researchers have focused on feature enhancement [33] and fusion [34], [35]. Zhang et al. [36] connected the output of each branch to the input of the next branch through feature reuse, enhancing the extraction and localization of detailed features. Duan et al. [37] proposed a spatial attention pyramid network (SAPN), which enhances the network's ability to represent multiscale features effectively. Wang et al. [38] achieved accurate classification and localization of remote sensing targets of varying sizes by constructing a multicorrelation auxiliary detection head. Although these methods have made some progress, they overlook the rich frequency information in high-resolution remote sensing images and the distortion or loss of target features caused by frequent upsampling and downsampling in complex feature fusion networks.

To build an efficient end-to-end object detection model for remote sensing images with wide-ranging variations in target sizes, this article proposes a scale adaptive and frequency fusion DETR (SAFF-DETR). The innovations of this study primarily manifest in key improvements tailored for the specific requirements of remote sensing object detection. Previous object detection methods primarily focused on spatial domain feature extraction and fusion, generally neglecting the abundant frequency information in remote sensing images. To bridge this gap, we leverage the multibranch representation fusion (MRF) module to fully exploit rich frequency features in shallow layers, thereby enhancing the detection performance of Transformer-based detectors for small objects. Moreover, conventional object detection methods typically employ repeated upsampling and downsampling operations for feature fusion. In contrast, we introduce a novel fusion architecture called cross-layer channelwise–spatialwise frequency fusion (CCSFF), which utilizes two carefully designed frequency-attention mechanisms for intragroup frequency interactions. This approach not only improves computational efficiency but also enhances multisize object representation, making it particularly suitable for remote sensing images with significant variations in target size.

Specifically, to enhance the detection capability of Transformer-based detectors for small targets, we construct the MRF module, which consists of a multibranch frequency enhancement (MFE) module and a frequency representation fusion (FR-Fusion) module, operating on shallow feature maps. This module exploits the rich frequency information of small targets, making shallow features more discriminative and effectively mitigating the information loss issue in small-target feature representation. Meanwhile, the CCSFF structure is introduced to integrate cross-layer frequency interactions between cross-layer spatial–frequency attention (CSFA) and cross-layer channel–frequency attention (CCFA), leveraging the Transformer's global modeling capabilities to achieve deep fusion of frequency domain information. This approach enables efficient multiscale feature interaction without relying on redundant upsampling and downsampling operations. Through the division of patch and the synergy of global modeling, frequency domain feature interactions are performed within each group, enabling the model to adapt the feature distributions of multisize targets and achieve accurate detection of small, medium, and large objects.

The main contributions of this article are listed as follows.

1) In order to address the issue of limited detection accuracy caused by large variations in target sizes, we propose an efficient end-to-end object detection algorithm for remote sensing images, achieving excellent

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performance for multisize object detection within a single framework.

2) In order to enhance the network's capability for small-target detection, the MRF, composed of MFE and FR-Fusion, is designed. Specifically, MFE exploits the rich frequency information of small targets in shallow features, while FR-Fusion further integrates the features extracted from multiple branches to generate more discriminative comprehensive features, boosting the network's ability to perceive small targets with minimal computational overhead.
3) In order to address the issue of small-target information loss and large target feature distortion caused by frequent upsampling and downsampling in traditional multiscale feature interaction steps, we introduce the CCSFF architecture, which integrates two frequency attention mechanisms (CSFA and CCFA) to achieve efficient multiscale frequency interaction in a single step, thereby enhancing the network's ability to perceive targets of varying sizes.

The remaining structure of this article is as follows. After discussing the related techniques in Section II, Section III describes the proposed methodology in detail. In Section IV, the experimental details are briefly described, and a number of experiments are conducted to validate the performance of the model. Section V provides a summary and a vision for future work.

II. Related Works

This section briefly reviews three key technologies related to our work: Transformer-based object detection methods, feature enhancement and multiscale fusion, and frequency domain analysis in vision.

A. Transformer-Based Object Detection Methods

The Transformer architecture was initially based on the attention mechanism and applied to NLP tasks. After Dosovitskiy et al. [39] applied it to the field of computer vision, the Vision Transformer (ViT) demonstrated significant potential, attracting extensive research from scholars [40], [41], [42]. In object detection tasks, DETR [15] and Pix2Seq [43] are among the earliest Transformer-based detectors, proposing two distinct paradigms for object detection. However, due to the quadratic computational complexity of the self-attention mechanism, its application to high-resolution feature maps faces certain limitations. To address this issue and enable the practical application of Transformers, Mehta and Rastegari [44], [45] and Wadekar and Chaurasia [46] proposed the lightweight MobileViT series, which achieves a strong balance between accuracy and real-time performance. Deformable DETR [16] integrates deformable convolutions and sparse attention mechanisms, reducing computational complexity and improving detection efficiency. O2-DETR [47] replaces attention modules with local convolutions and integrates multiscale feature maps, demonstrating excellent detection performance in scenarios involving rotated objects. DINO [19] optimizes the efficiency and performance of previous DETR-like models by employing a contrastive denoising training approach, along with an anchor-initialized hybrid query selection method and a "look forward twice" scheme for box prediction. RT-DETR [32] achieves real-time object detection by reducing the complexity of the Transformer architecture and minimizing the input sequence length, surpassing mainstream real-time detection models like YOLO in both speed and accuracy for the first time.

B. Feature Enhancement and Multiscale Fusion

Feature enhancement and multiscale fusion aim to represent the same target at different scales, combining detail-rich shallow features with semantically strong deep features to improve the semantic representation capability of the features. Luo et al. [48] demonstrated the importance of enhancing semantic information in features for object detection. Therefore, in the context of remote sensing images, feature enhancement and multiscale fusion enable better interpretation of background information and target relationships, enhancing the detection accuracy of multisize objects.

Lin et al. [49] proposed the feature pyramid network (FPN), which enhances semantic information in shallow feature maps through top-down feature fusion. Building on FPN, Liu et al. [50] proposed PAFPN, which supplements the missing shallow information in top-level feature maps through a bottom-up branch. Feng et al. [51] introduced a branched gate unit structure in the backbone network to capture multiscale feature information, improving the performance of ship detection in SAR images. Yang et al. [52] introduced AFPN, which refines multiscale features through iterative cross-layer fusion of deep and shallow feature maps, demonstrating excellent performance in object detection scenarios with common scale distributions. Liu et al. [53] proposed a method combining a spatial context pyramid module and a hierarchical region-of-interest extraction module, enhancing the model's ability to extract multiscale features of the same target, thereby providing more accurate and comprehensive feature representations for remote sensing object detection. However, existing multiscale feature fusion modules still rely on traditional upsampling and downsampling operations to align the resolutions of feature maps from different layers, which often leads to the loss of shallow-level detail features and the distortion of deep-level features.

C. Frequency Domain Analysis in Vision

In the field of vision, researchers have conducted extensive studies on the application of frequency domain analysis. Related studies [54], [55] have shown that low-frequency components in images primarily capture global structures and color information, while high-frequency components contain detailed features of targets, such as sharp edges. This finding has been widely applied in image super-resolution [56], image recompression [57], and generalization [58]. Pan et al. [59] proposed HiLo attention, which extracts local fine details and global structures through separate high- and low-frequency branches, reducing complexity and computational

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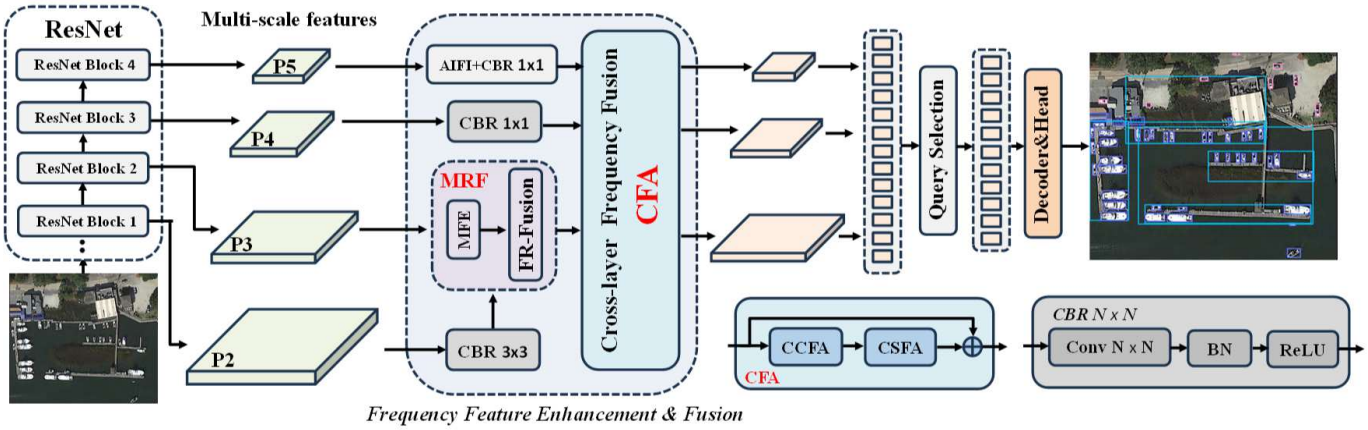


Fig. 2. Overall framework of SAFF-DETR. First, remote sensing images are fed into the backbone network. Then, frequency feature enhancement and fusion are performed on the extracted multiscale feature maps at the neck. In this stage, the AIFI module serves as the intrascale feature interaction component, whose architecture aligns with that of a standard Transformer encoder. Finally, the interacted multiscale features are used in the head to predict target categories and bounding boxes.

costs compared with traditional global attention mechanisms. High-resolution remote sensing images contain rich frequency features, but these are often overlooked, resulting in insufficient representation of objects with varying sizes.

III. Proposed Method

Existing object detection models often struggle to accurately detect targets of varying sizes within a single framework. To address this issue, we propose a novel approach called SAFF-DETR. As illustrated in Fig. 2, SAFF-DETR is an efficient end-to-end detector specifically designed for multisize object detection tasks in remote sensing imagery: 1) to fully exploit shallow frequency features and enhance the representation capability of small targets without significantly increasing computational costs, we utilize two lightweight modules, MFE and FR-Fusion, for efficient multibranch frequency domain feature representation and fusion; 2) by applying CCFA and CSFA to multiscale feature fusion, cross-layer frequency fusion is performed along the spatial and channel dimensions, further enhancing the semantic representation capability of features and suppressing background noise; and 3) by replacing the traditional feature pyramid structure with CCSFF, we achieve one-step multiscale feature fusion, effectively avoiding feature loss of small targets and distortion of large ones caused by upsampling and downsampling, while enhancing the model's capability to detect targets of varying sizes.

A. MRF Module

The complex backgrounds in remote sensing images and the limited feature extraction capability of the backbone make it difficult for detectors to effectively distinguish small targets from the background. To overcome the inherent limitation of Transformer-based detectors, which exhibit strong perception for large targets but insufficient detection capability for small ones, we design an MRF module aimed at enhancing the network's ability to detect small targets. Shallow feature maps, with their higher resolution, contain rich positional and detailed information, which is crucial for small-target detection. Therefore, traditional methods for improving small-target detection often introduce an additional shallow detection layer, such as the P2 layer. However, this approach typically results in a significant increase in computational cost and longer processing times. To solve these issues, we propose a strategy that balances efficiency and performance. We place the MRF module before the multiscale feature interaction and adopt a serial fusion structure. By deeply integrating the P2-layer and P3-layer features as the input to the MRF, the method fully leverages shallow features while minimizing computational overhead.

The shallow features rich in small-target information are simply concatenated with the feature layer at a higher level of resolution, serving as the input to the MRF module, denoted as $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$. As shown in Fig. 3(a), the MFE module effectively reduces computational costs by incorporating a cross-stage partial (CSP) structure. Its design includes three branches: a local branch, a large branch, and a frequency enhancement branch, aimed at enhancing multiscale representation. Each branch extracts feature information at different levels and completes feature fusion through addition operations. The fused features are modulated by a 1×1 convolution and then concatenated with the output of the bypass branch to enrich the final feature representation. In the local branches, we utilize a simple 1×1 depthwise convolutional layer to enhance local feature representation. While in the large branch, we introduce a 31×31 depthwise convolution to capture a larger receptive field, thereby extracting more global information. In addition, 31×1 and 1×31 depthwise convolutions are designed in parallel with the square convolution to efficiently extract strip-shaped contextual information. This combination enables modeling in both horizontal and vertical directions, thereby enhancing the perception of fine-grained frequency features. In the frequency enhancement branch, we further enhance the representation capability of frequency features by introducing a dual-domain processing mechanism. Specifically, this branch integrates spatial and channel domain feature enhancement modules, namely, channel–frequency enhancement (CFE) and spatial–frequency

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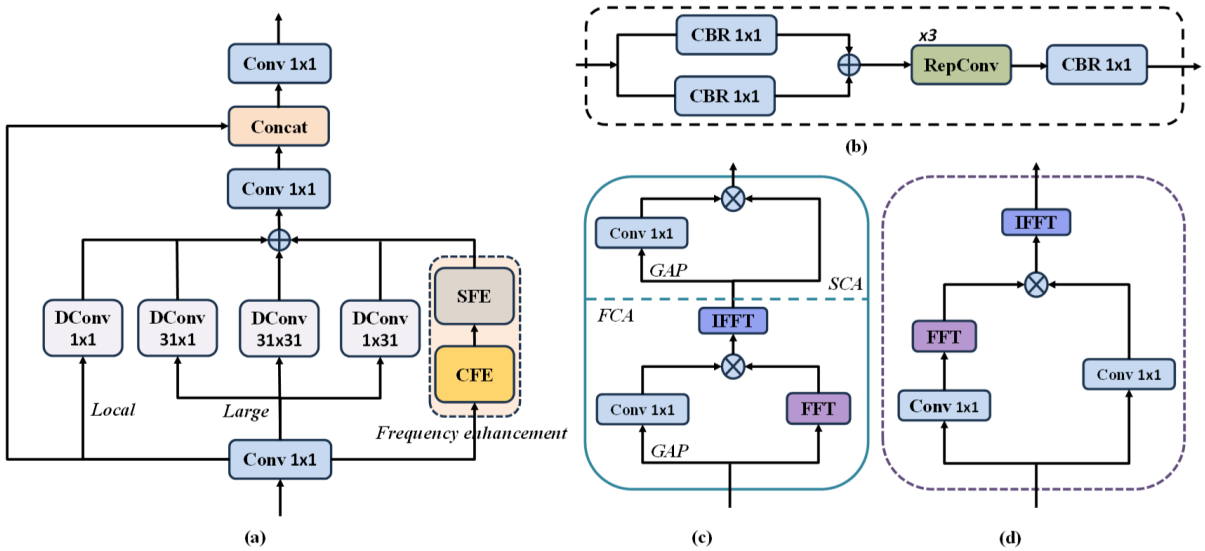


Fig. 3. MFE and FR-Fusion simply connected in series to form MRF. (a) MFE. (b) FR-Fusion. (c) CFE. (d) SFE.

enhancement (SFE), which optimize frequency feature in the channel and spatial domains, respectively, as shown in Fig. 3(c) and (d). For the input feature $\mathbf{X_{FE}} \in \mathbb{R}^{C' \times H \times W}$ of the frequency enhancement branch, CFE first applies frequency channel attention (FCA) to $\mathbf{X_{FE}}$ as follows:

$$\mathbf{X}_{\mathrm{FCA}} = \mathcal{IF}\left(\mathcal{F}\left(\mathbf{X_{FE}}\right) \otimes \mathbf{W}_{\mathbf{1\times 1}}^{\mathrm{FCA}}\left(\mathrm{GAP}\left(\mathbf{X_{FE}}\right)\right)\right) \tag{1}$$

where $\mathcal{F}$ and $\mathcal{IF}$ denote the Fourier and inverse Fourier transforms, respectively, $\mathbf{X}_{\mathrm{FCA}}$, $\mathbf{W_{1\times 1}}$, and GAP denote the output of FCA, 1×1 convolution, and global average pooling, respectively, and $\otimes$ denotes an element-by-element multiplication operation. The channel–frequency features of the image are effectively refined through the Fourier transform. After the channel spectrum modulation, the obtained features are further sent to the spatial channel attention (SCA) module, which can be formally expressed as follows:

$$\mathbf{X}_{\mathrm{CFE}} = \mathbf{X}_{\mathrm{FCA}} \otimes \mathbf{W}_{\mathbf{1\times 1}}^{\mathrm{SCA}}\left(\mathrm{GAP}\left(\mathbf{X}_{\mathrm{FCA}}\right)\right) \tag{2}$$

where $\mathbf{X}_{\mathrm{CFE}}$ is the output of CFE, which enhances frequency features along the channel dimension. Subsequently, we apply a frequency-based attention module in the spatial dimension to strengthen frequency features along the spatial direction, formulated as

$$\mathbf{X}_{\mathrm{SFE}} = \mathcal{IF}\left(\mathcal{F}\left(\mathbf{W}_{\mathbf{1\times 1}}^{\mathbf{1}}\left(\mathbf{X}_{\mathrm{CFE}}\right)\right) \otimes \mathbf{W}_{\mathbf{1\times 1}}^{\mathbf{2}}\left(\mathbf{X}_{\mathrm{CFE}}\right)\right) \tag{3}$$

where $\mathbf{X}_{\mathrm{SFE}}$ is the result after applying SFE. By sequentially combining CFE and SFE, the model focuses on frequency information that helps separate small targets from the background. As shown in Fig. 3(b), the FR-Fusion module employs a lightweight reparameterized convolution structure to efficiently integrate the multibranch features extracted by the MFE module, generating a comprehensive feature representation with discriminative capability. In Section IV, we evaluate the network's capability for small-target detection on the VEDAI dataset.

B. Cross-Layer Frequency Attention (CFA)

Through CFA, the model enhances its ability to leverage global contextual information and frequency features. Specifically, CFA is composed of CCFA and CSFA. CCFA establishes a global receptive field in the spatial dimension through cross-layer interactions along the channel dimension, based on intragroup high- and low-frequency information. CSFA, on the other hand, performs cross-layer interactions along the spatial dimension and establishes a global receptive field in the channel direction according to high and low frequencies. Finally, the gradient gain is further improved through residual connections.

Assume that each multiscale feature map input to the cross-layer frequency fusion can be represented as $\mathbf{X} = \left\{\mathbf{X_i} \in \mathbb{R}^{C\times H_i\times W_i}\right\}_{i=1}^{L}$, where L is the number of input layers. The spatial resolution $H_i \times W_i$ of the features of each layer increases with i, but the number of channels C remains constant. CFA can be expressed as follows:

$$Y = \mathrm{CFA}\left(\mathbf{X}\right) = \mathrm{CSFA}\left(\mathrm{CCFA}\left(\mathbf{X}\right)\right) + \mathbf{X} \tag{4}$$

where $\mathbf{Y} = \left\{\mathbf{Y_i} \in \mathbb{R}^{C\times H_i\times W_i}\right\}_{i=1}^{L}$ is the multiscale feature map after cross-layer frequency interaction, maintaining the same shape as the input.

1) Cross-Layer Channelwise Frequency Attention: CCFA performs cross-layer frequency interactions along the channel dimension, providing global contextual information in the spatial dimension for each channel token. For the input set of feature maps $\mathbf{X} = \left\{\mathbf{X_i} \in \mathbb{R}^{C\times H_i\times W_i}\right\}_{i=1}^{L}$, we first perform a spatial alignment (SA) operation. With this operation, the feature maps on each scale are reconstructed so that their spatial resolution is consistent, thus obtaining the aligned feature set $\hat{\mathbf{X}} = \left\{\hat{\mathbf{X}}_\mathbf{i} \in \mathbb{R}^{(H_iW_i/H_1W_1)\times C\times H_1\times W_1}\right\}_{i=1}^{L}$. Specifically, the SA operation stacks the spatial dimension features along the channel direction, which ensures that the spatial information of the features can be expressed at a consistent scale while

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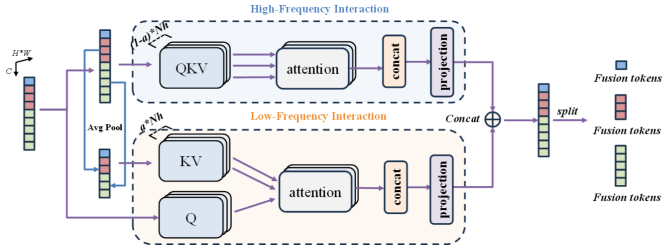


Fig. 4. Framework of CCFA. N_h represents the number of heads in the multihead attention. α denotes the proportion of high- and low-frequency feature interaction heads.

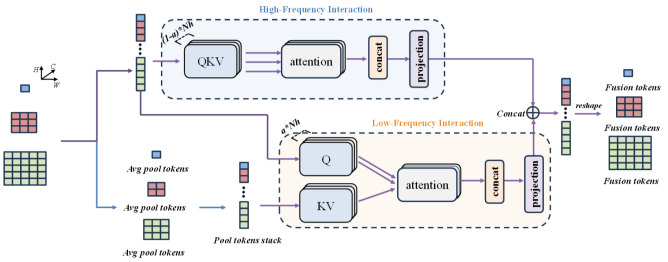


Fig. 5. Framework of CSFA. The meaning of N_h and α is the same as in CCFA.

maintaining computational efficiency. The above operation can be expressed as follows:

$$\hat{\mathbf{X}} = \mathrm{SA}(\mathbf{X}). \tag{5}$$

Considering the channel direction as the number of tokens, each scale will generate $T_c = \{\mathbf{T_i} \in (H_i \mathrm{W}_i / H_1 W_1) \times C\}_{i=1}^{L}$ tokens, and $H_1 \times W_1$ is the dimension of each token. The tokens from each scale are concatenated to obtain $\bar{\mathbf{X}}_\mathbf{c} \in \mathbb{R}^{\sum_{i=1}^{L}(H_i W_i / H_1 W_1) \times C, H_1 \times W_1}$, which is used as the input to the CCFA. The CCFA process is shown in Fig. 4. Specifically, we divide the heads of multihead attention into two groups with a ratio of α, where $(1-\alpha) N_h$ heads are used for high-frequency interactions and the remaining αN_h heads are used for low-frequency interactions. In the high-frequency interaction part, each head utilizes linear transformations of $\bar{\mathbf{X}}_\mathbf{c}$ to compute the query $\mathbf{Q}$, key $\mathbf{K_h}$, and value $\mathbf{V_h}$

$$\mathbf{Q} = \bar{\mathbf{X}}_\mathbf{c} \mathbf{W_q}, \quad \mathbf{K_h} = \bar{\mathbf{X}}_\mathbf{c} \mathbf{W_k^h}, \quad \mathbf{V_h} = \bar{\mathbf{X}}_\mathbf{c} \mathbf{W_v^h} \tag{6}$$

where $\mathbf{W_q}, \mathbf{W_k^h}, \mathbf{W_v^h} \in \mathbb{R}^{H_1 \times W_1 \times D_h}$ is the learnable parameter and D_h is the hidden dimension of each head. Subsequently, the output of each head is computed using the scaled dot product and concatenated with multiple heads, denoted as follows:

$$\mathrm{SA}_h(\bar{\mathbf{X}}_c) = \mathrm{Softmax}\left(\frac{\mathrm{QK_h^\top}}{\sqrt{D_h}} + P\right)\mathbf{V}_h \tag{7}$$

$$\mathrm{MSAH}(\bar{\mathbf{X}}_c) = \underset{h \in [(1-\alpha)N_h]}{\mathrm{concat}} \left[\mathrm{SA}_h(\bar{\mathbf{X}}_c)\right] \mathbf{W}_{\mathrm{oh}}^c \tag{8}$$

where $\mathrm{SA}_h(\bar{\mathbf{X}}_c)$ and P represent the output of a single head and the positional encoding in high-frequency interactions, respectively. $\mathrm{MSAH}(\bar{\mathbf{X}}_c)$ is the output of high-frequency interaction, and $\mathbf{W}_{\mathrm{oh}}^c \in \mathbb{R}^{(1-\alpha)N_h \mathrm{D}_h \times (1-\alpha) H_1 W_1}$ is a learnable parameter. In the low-frequency interaction part, since averaging acts as a low-pass filter [60], we first apply average pooling to the input to extract the low-frequency signal $\bar{\mathbf{X}}_\mathbf{c}^\mathbf{l}$ from $\bar{\mathbf{X}}_\mathbf{c}$, and then map the averaged features to $\mathbf{K_l}, \mathbf{V_l} \in \mathbb{R}^{\sum_{i=1}^{L}(H_i W_i / H_1 W_1 s) \times C \times D_h}$, where s is the average window size. The multihead attention is computed using the scaled dot product based on $\mathbf{Q}, \mathbf{K_l}$, and $\mathbf{V_l}$, as follows:

$$\mathrm{SA}_l(\bar{\mathbf{X}}_c) = \mathrm{Softmax}\left(\frac{\mathrm{QK_l^\top}}{\sqrt{D_h}} + P\right)\mathbf{V}_l \tag{9}$$

$$\mathrm{MSAL}(\bar{\mathbf{X}}_c) = \underset{h \in [\alpha N_h]}{\mathrm{concat}} \left[\mathrm{SA}_l(\bar{\mathbf{X}}_c)\right] \mathbf{W}_{\mathrm{ol}}^c \tag{10}$$

where $\mathbf{W}_{\mathrm{ol}}^c \in \mathbb{R}^{\alpha N_h \mathrm{D}_h \times \alpha H_1 W_1}$ is a learnable parameter. The final CCFA output is concatenated from two branches

$$\mathrm{CCFA}(\bar{\mathbf{X}}_c) = \left[\mathrm{MSAH}(\bar{\mathbf{X}}_c); \mathrm{MSAL}(\bar{\mathbf{X}}_c)\right]. \tag{11}$$

TABLE I
ABLATION EXPERIMENTS ON NWPU VHR-10 DATASET

Baseline	MRF	CCFA	CSFA	$\mathrm{mAP}_{50}(\%)$	$\mathrm{mAP}_{50:95}(\%)$	Params(M)
✓				92.1	60.8	19.88
✓	✓			93.0	61.1	21.46
✓		✓		92.6	60.4	18.55
✓			✓	92.5	59.0	**18.30**
✓		✓	✓	93.2	59.7	18.66
✓	✓	✓	✓	**93.5**	**61.8**	20.60

TABLE II
IMPACT OF DIFFERENT MRF DEPLOYMENT STRATEGIES ON MODEL PERFORMANCE

P2	P3	P4	P5	$\mathrm{mAP}_{50}(\%)$	$\mathrm{mAP}_{50:95}(\%)$	GFlops
	✓			91.9	59.9	63.8
		✓		92.5	61.1	56.6
			✓	92.3	60.9	**54.8**
✓	✓			**93.5**	**61.8**	78.5

$[\bullet]$ denotes a concatenate operation.

2) Cross-Layer Spatialwise Frequency Attention: As shown in Fig. 5, CSFA performs cross-layer frequency interactions along the spatial dimension, providing global contextual information in the channel dimension for each spatial token. Similarly, let the input feature map set for CSFA be $\mathbf{X} = \{\mathbf{X_i} \in \mathbb{R}^{C \times H_i \times W_i}\}_{i=1}^{L}$. Since the channel dimensions of $\mathbf{X}$ are already aligned, looking at the spatial direction as the number of tokens, $\mathbf{T_s} = \{\mathbf{T_i} \in H_i \mathrm{W}_i\}_{i=1}^{L}$ tokens will be generated per scale. Concatenate the tokens from each scale to obtain $\bar{\mathbf{X}}_\mathbf{s} \in \mathbb{R}^{\sum_{i=1}^{L} H_i W_i \times C}$. Use $\bar{\mathbf{X}}_\mathbf{s}$ to linearly transform into $\mathbf{Q}, \mathbf{K_h}, \mathbf{V_h} \in \mathbb{R}^{\sum_{i=1}^{L} \mathrm{H_i W_i} \times D_h}$, then compute the output of a single head using the scaled dot product and concatenate multiple heads, as shown in the following:

$$\mathrm{SA}_h(\bar{\mathbf{X}}_s) = \mathrm{Softmax}\left(\frac{\mathrm{QK_h^\top}}{\sqrt{D_h}} + P\right)\mathbf{V}_h \tag{12}$$

$$\mathrm{MSAH}(\bar{\mathbf{X}}_s) = \underset{h \in [(1-\alpha)N_h]}{\mathrm{concat}} \left[\mathrm{SA}_h(\bar{\mathbf{X}}_s)\right] \mathbf{W}_{\mathrm{oh}}^s \tag{13}$$

where $\mathbf{W}_{\mathrm{oh}}^s \in \mathbb{R}^{(1-\alpha)N_h \mathrm{D}_h \times (1-\alpha) C}$ is a learnable parameter. In the low-frequency interaction part, average pooling is similarly applied to the input to extract the low-frequency signal $\bar{\mathbf{X}}_\mathbf{l}$ from $\mathbf{X}$, followed by token concatenation along the spatial dimension to obtain $\bar{\mathbf{X}}_\mathbf{s}^\mathbf{l} \in \mathbb{R}^{\sum_{i=1}^{L}(H_i W_i / s^2) \times C}$. Use $\bar{\mathbf{X}}_\mathbf{s}^\mathbf{l}$ to map the averaged features to $\mathbf{K_l}, \mathbf{V_l} \in \mathbb{R}^{\sum_{i=1}^{L}(H_i \mathrm{W}_i / s^2) \times D_h}$. Similarly, multihead attention is computed using the scaled dot product

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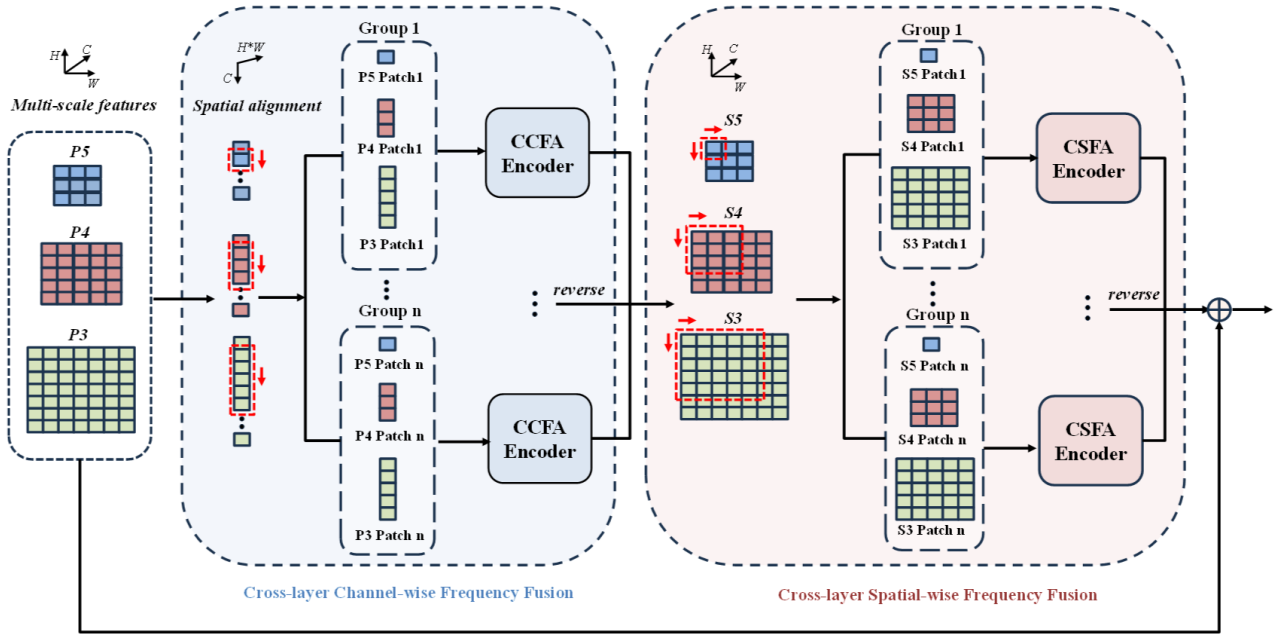


Fig. 6. CCSFF structure. The multiscale feature map is first operated by SA, and the channel direction cross-layer frequency interaction is carried out in each group. The output obtained is then turned back to the spatial direction, and the spatial direction cross-layer frequency interaction is carried out in each group.

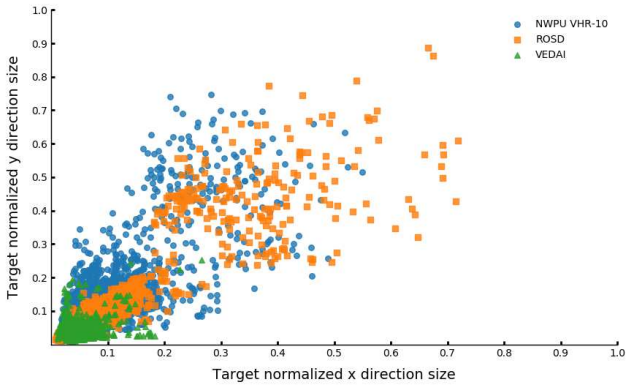


Fig. 7. Object size distribution in three datasets. Closer to the lower left corner indicates a smaller target size, and the upper right corner indicates a larger size.

based on $\mathbf{Q}, \mathbf{K_l}$, and $\mathbf{V_l}$, as follows:

$$\mathrm{SA}_l(\mathbf{X}) = \mathrm{Softmax}\left(\frac{\mathrm{QK}_\mathbf{l}^\top}{\sqrt{D_h}} + P\right)\mathbf{V}_l \tag{14}$$

$$\mathrm{MSAL}(\mathbf{X}) = \underset{h\in[\alpha N_h]}{\mathrm{concat}}\left[\mathrm{SA}_l(\mathbf{X})\right]\mathbf{W}_{\mathrm{ol}}^s \tag{15}$$

where $\mathbf{W}_{\mathrm{ol}}^s \in \mathbb{R}^{\alpha N_h \mathrm{D}_h \times \alpha C}$ is the learning parameter. The final CSFA output is represented as

$$\mathrm{CSFA}\left(\bar{\mathbf{X}}_s, \mathbf{X}\right) = \left[\mathrm{MSAH}\left(\bar{\mathbf{X}}_s\right); \mathrm{MSAL}(\mathbf{X})\right]. \tag{16}$$

C. CCSFF Structure

Traditional multiscale feature fusion steps involve complex upsampling and downsampling operations and rely on layer-by-layer information transmission mechanisms, often leading to information loss and increased memory access latency. By utilizing the efficient global modeling of CCFA and CSFA, we introduce the CCSFF structure for single-step cross-layer interactions without distance constraints. Given that the computational complexity of the Transformer encoder grows quadratically with sequence length [14], inspired by Swin Transformer [42], as shown in Fig. 6, we perform overlapping window operations within multiscale feature maps and conduct frequency interactions within each group. The small-target redundancy information generated by overlapping sliding windows is transformed into a beneficial feature enhancement mechanism through CCSFF, while achieving a significant reduction in overall computational complexity to $O(N)$ level. Specifically, after performing the SA operation on the input feature map set $\mathbf{X} = \left\{\mathbf{X_i} \in \mathbb{R}^{C\times H_i\times W_i}\right\}_{i=1}^{L}$, we obtain $\hat{\mathbf{X}} = \left\{\hat{\mathbf{X}}_\mathbf{i} \in \mathbb{R}^{(H_iW_i/H_1W_1)\times C\times H_1\times W_1}\right\}_{i=1}^{L}$. Subsequently, we design the sliding window size based on the shape of $\hat{\mathbf{X}}$ to ensure that the number of patches obtained after overlapping window operations is consistent across different scales. Assume $\hat{\mathbf{X}}_{\mathrm{group}}^{\mathbf{C}} = \left\{\hat{\mathbf{X}}_{\mathrm{group}}^{\mathbf{i}} \in \mathbb{R}^{S_i\times H_1\times W_1}\right\}_{i=1}^{L}$ is the set of patches obtained from sliding window operations at different scales. S_i is the window size on different scales. The $\hat{\mathbf{X}}_{\mathrm{group}}^{\mathbf{C}}$ is treated as a group and used as input to CCFA for modeling intragroup frequency interactions along the channel dimension. Similarly, the operation of the overlapping window in the spatial dimension is analogous to the previous step. For input $\mathbf{X} = \left\{\mathbf{X_i} \in \mathbb{R}^{C\times H_i\times W_i}\right\}_{i=1}^{L}$, overlapping window operations are performed directly on the basis of the size of the multiscale feature maps, resulting in $\hat{\mathbf{X}}_{\mathrm{group}}^{\mathbf{S}} = \left\{\hat{\mathbf{X}}_{\mathrm{group}}^{\mathbf{i}} \in \mathbb{R}^{S_i\times S_i\times C}\right\}_{i=1}^{L}$. It is then used as the input to CSFA for intragroup frequency interactions along the spatial dimension. Benefiting from the division strategy for multiscale patches, high-resolution shallow features can participate more extensively in feature interactions across different groups, thereby further enhancing the representation capability of shallow detail features. Furthermore, by introducing cross-layer consistent relative

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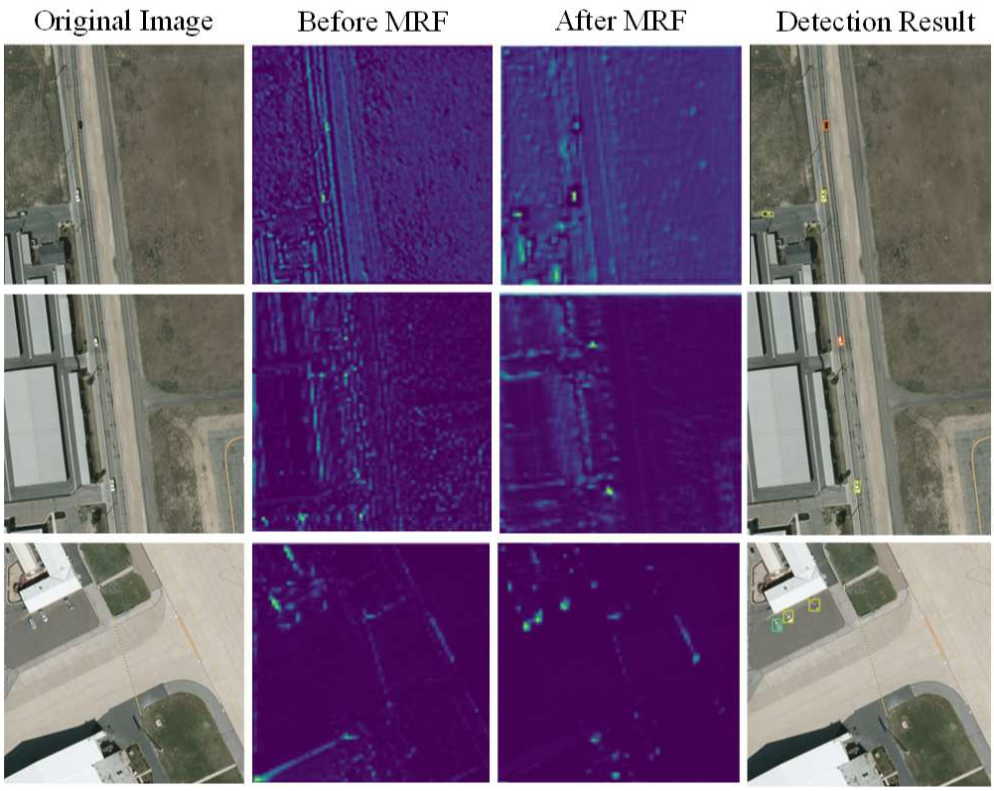


Fig. 8. Effect of MRF, where the highlights represent the areas of focus of the model.

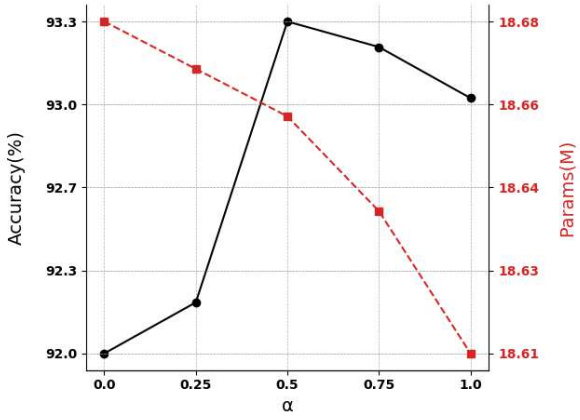


Fig. 9. Impact of α on the network, the larger of α, the more low-frequency interactions.

positional encoding, the perception of cross-layer positional information during feature interactions in CCSFF is further improved, significantly enhancing the feature fusion effect.

IV. Experiments and Results

To thoroughly validate the detection performance of the proposed model for remote sensing images with significant target size variations, we evaluate and analyze the model's performance on two public remote sensing datasets with diverse target size distributions, NWPU VHR-10 and RSOD, as well as the small-target remote sensing dataset VEDAI. The size distributions of targets in each dataset are shown in Fig. 7. As can be seen, the NWPU VHR-10 and RSOD datasets exhibit a wide range of target size variations, while the targets in the VEDAI dataset are concentrated in smaller size ranges.

A. Datasets

1) NWPU VHR-10 Dataset [61]: The NWPU VHR-10 dataset is an optical remote sensing dataset released by the Northwestern Polytechnical University, which includes 800 ultrahigh resolution (VHR) remote sensing images from Google Earth and Vaihingen databases. The image resolution ranges from 0.5 to 2.0 m and covers a wide range of geographic regions and scene environments. A total of 3651 target instances were labeled, covering ten categories with large-size differences. Specifically including airplane (AP), ship (SP), storage tank (ST), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GF), harbor (HB), bridge (BR), and vehicle (VE).

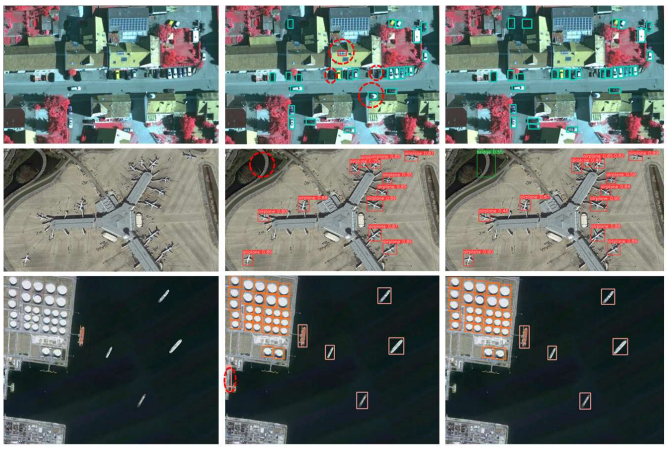
2) RSOD Dataset [62]: The RSOD dataset is a high-resolution dataset designed for remote sensing image object detection tasks. It contains 976 high-resolution optical remote sensing images with a total of 6950 instances, covering four common remote sensing target categories: aircraft (AC), oil tanks (OTs), overpasses (OPs), and playgrounds (PGs). Each image has a resolution of 800 $\times$ 800.

3) VEDAI Dataset [63]: VEDAI is a public small-target dataset consisting of satellite imagery taken in Utah in 2012. It contains 1210 pairs of RGB and IR images of the same scene

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TABLE III
COMPARISON RESULTS ON NWPU VHR-10 DATASET

	mAP_{50}	$mAP_{50:95}$	AP	SP	ST	BD	TC	BC	GF	HB	BR	VE	Params(M)	GFlops
YOLOv8-M [64]	90.8	56.3	99.1	88.8	91.7	98.7	83.3	91.3	99.2	85.9	91.8	88.7	25.8	78.7
YOLOv10-L [65]	87.6	56.1	99.2	86.3	87.2	96.0	86.8	84.0	99.0	83.4	84.2	87.3	25.7	126.4
YOLOX-M [66]	88.2	54.7	98.2	75.4	87.6	98.8	88.9	86.5	99.8	65.8	99.4	84.1	25.3	36.8
FFCA-YOLO [33]	90.6	54.9	99.2	85.6	93.1	98.7	87.5	94.3	99.2	**89.7**	75.6	86.6	7.2	52.0
TOOD [67]	83.7	50.6	98.9	77.0	88.9	96.4	78.7	85.1	94.2	63.1	72.8	81.8	32.0	195.5
FCOS [68]	91.7	54.9	98.9	**90.6**	93.7	95.6	93.6	92.6	99.9	79.1	82.3	90.7	19.1	150.5
RT-DETR [32]	91.7	60.4	99.1	83.2	**94.9**	97.9	95.4	86.3	99.0	80.6	95.1	89.0	19.9	57.0
DINO [19]	92.0	59.0	99.2	78.3	92.9	95.5	89.9	92.0	**100.0**	85.3	94.5	88.2	47.5	265.0
DDQ [69]	90.3	57.4	98.6	88.5	96.4	96.8	89.7	92.6	99.9	76.6	76.0	88.1	43.7	98.6
Lightweight Detectors														
FastestDet [70]	74.1	38.1	88.2	70.2	82.4	80.2	73.5	82.7	90.3	63.8	58.6	72.0	**0.24**	**1.0**
YOLOX-S [66]	87.2	53.1	98.1	81.9	88.2	98.2	85.5	88.6	99.9	64.2	88.8	77.4	8.94	13.4
YOLOv8-S [64]	90.1	56.1	98.5	88.6	81.4	98.0	82.8	89.3	99.1	89.0	**99.5**	91.0	11.3	28.5
YOLOv5-Lite [71]	90.2	57.6	98.6	84.4	89.7	96.3	88.3	94.0	99.5	77.9	89.5	**91.2**	5.3	15.2
Multi-size Object Detector (ours)														
SAFF-DETR	**93.3**	**61.5**	**99.3**	81.9	92.2	**98.9**	**96.0**	**94.3**	99.4	87.3	98.6	88.1	20.6	78.5



(a) (b) (c)

Fig. 10. Comparison of target detection results. The red dotted box indicates a missed or false check. (a) Original images. (b) Baseline detection results. (c) Our detection results.

with a resolution of 1024 × 1024, and we only experimented on RGB images. The data contains nine classes of small-target objects, namely, boat (BT), camping car (CC), car (CA), pick up (PU), plane (PA), tractor (TA), truck (TR), van (VA), and VE, for a total of 1274 instances. Notably, these targets are not only small in size but also exhibit considerable diversity and complexity, such as variations in orientation, lighting, shadows, and target occlusion, further increasing the detection difficulty.

The experimental study employs three benchmark datasets that cover diverse geographical types, including urban areas, ports, airports, agricultural regions, and deserts, with a combined coverage extending across more than seven major geographic divisions. These datasets exhibit remarkable data diversity and feature a continuous spectrum of image resolutions, ranging from ultrahigh 0.3-m to medium 2-m spatial resolution. Moreover, the target sizes form a complete dimensional spectrum ranging from small-scale objects averaging 28 × 28 pixels (e.g., vehicles in VEDAI) to large-scale targets exceeding 256 × 256 pixels (e.g., airport runways in NWPU VHR-10). This comprehensive scale coverage establishes a rigorous validation foundation for assessing object detection algorithms' generalization performance across varying target scales.

B. Experimental Environment and Settings

In this article, all experiments are implemented based on the PyTorch framework. The experimental running environment is as follows: GPU is NVIDIA GeForce RTX 4090, CPU is Intel[1]Xeon[1] Gold 6271C at 2.60 GHz, and Linux OS. The optimizer, initial learning rate, momentum, and weight decay during training are AdamW, 0.0001, 0.9, and 0.0001, respectively. For the experiments, 80% of the images from NWPU VHR-10 and ROSD were randomly selected as the training set and the remaining 20% as the testing set. VEDAI is divided into training, validation, and testing sets by 8:1:1.

C. Ablation Experiment

To evaluate the impact of each module and parameter in our proposed SAFF-DETR algorithm on model performance, we conduct ablation experiments on the NWPU VHR-10 dataset. Assessment metrics included mAP_{50}, $mAP_{50:95}$, and model parameters. Table I lists the experimental results obtained by SAFF-DETR under various optimization measures.

1) Multibranch Representation Fusion: This module adopts a multibranch structure design, which effectively captures multiscale spatial features in shallow feature maps. Meanwhile, the Fourier transform is introduced to extract detailed frequency features. Through the collaborative representation of spatial and frequency domain features, the discriminative capability for small objects is significantly enhanced. As shown in Table I, the addition of the MRF module improved all metrics, with mAP_{50} improving from 92.1% to 93.0% and $mAP_{50:95}$ improved from 60.8% to 61.1%, which fully

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TABLE IV
COMPARISON RESULTS ON RSOD DATASET

	mAP_{50}	$mAP_{50:95}$	mAP_S	mAP_M	mAP_L	AC	OT	OP	PG
YOLOV8-M [64]	94.9	67.2	44.7	69.7	72.1	96.6	98.8	84.7	98.8
YOLOV10-L [65]	92.5	65.3	44.7	69.0	70.1	96.4	98.5	80.8	96.6
YOLOX-M [66]	89.5	58.1	42.3	70.0	62.3	95.4	98.8	65.6	98.1
FFCA-YOLO [33]	90.3	61.2	42.9	69.0	66.1	96.7	98.4	73.7	93.7
TOOD [67]	87.8	52.5	32.1	62.5	57.6	92.1	97.2	67.0	95.1
FCOS [68]	94.4	61.7	38.7	66.2	66.5	94.3	97.6	85.7	99.8
RT-DETR [32]	95.0	68.0	44.5	72.4	72.5	96.6	99.0	85.3	99.3
DDQ [69]	95.1	67.4	39.9	69.3	72.5	95.4	98.8	**94.0**	**99.9**
DINO [19]	94.8	67.2	40.6	67.8	72.4	92.0	95.9	91.4	99.8
SAFF-DETR	**95.3**	**69.3**	**47.7**	**72.8**	**74.0**	**96.8**	**99.1**	85.7	99.5

TABLE V
COMPARISON RESULTS ON VEDAI DATASET

	mAP_{50}	$mAP_{50:95}$	BT	CC	CA	PU	PA	TA	TR	VA	VE
YOLOV8-M [64]	43.1	24.1	29.2	34.8	47.2	52.9	99.4	64.7	28.7	39.7	**37.9**
YOLOV10-L [65]	38.4	22.3	29.1	34.1	35.7	40.4	99.3	58.8	22.5	40.1	16.8
YOLOX-M [66]	24.4	13.6	15.1	14.0	19.2	26.2	10.1	19.9	13.1	15.4	3.0
TOOD [67]	40.0	22.1	20.6	29.4	20.9	28.5	11.8	30.7	20.3	37.6	21.3
DINO [19]	48.4	28.9	30.3	32.1	35.0	36.8	40.3	31.6	34.8	34.8	12.9
RT-DETR [32]	44.9	25.7	41.6	**58.9**	55.5	53.2	99.3	64.3	19.3	25.6	25.0
RTMDET [72]	47.8	29.6	26.2	30.2	31.2	40.7	33.4	38.5	22.9	50.6	22.2
Small Object Detectors											
FFCA-YOLO [33]	40.3	22.0	39.8	39.6	46.6	**54.7**	69.5	64.4	28.8	44.2	15.8
UAV-DETR [73]	46.1	27.0	41.5	51.0	42.3	50.2	82.8	**75.6**	36.2	51.4	36.3
SR-TOD [74]	48.1	22.3	25.1	31.1	26.6	32.2	14.2	26.3	17.1	33.9	16.5
Multi-size Object Detector (ours)											
SAFF-DETR	**48.7**	**29.8**	**44.1**	56.5	**55.8**	54.1	**99.5**	66.2	**38.7**	**51.6**	29.9

validates the effectiveness of the MRF module. Meanwhile, we conducted a comprehensive study on the impact of different MRF configuration strategies on model performance, with the results summarized in Table II. Although our adopted strategy incurs some computational overhead, it achieves significant accuracy improvements compared with alternative approaches. To further validate the effectiveness of MRF in enhancing small object detection, we visualized the feature maps before and after MRF processing, as shown in Fig. 8. Bright areas in the feature map indicate that the model pays more attention to that area. As can be seen from the figure, after adding the MRF module, the model can distinguish the small targets from the background more clearly, thus improving the detection performance.

2) Cross-Layer Channel–Frequency Attention: Building upon the frequency features extracted by MRF, CCFA optimizes feature fusion efficiency through an intragroup channel token frequency interaction mechanism. By adaptively fusing frequency characteristics across different channel tokens, it enhances the model's semantic discrimination capability between targets and background. The addition of CCFA to the baseline network enhances the model's ability to capture global contextual information along the spatial dimension. As shown in Table I, mAP_{50} improved from 92.1% to 92.6%, and the number of model parameters decreased from 19.88M to 18.55M. Experimental results show that the cross-layer channel–frequency fusion attention mechanism performs more efficiently in multiscale feature fusion. Adopting a high- and low-frequency interaction strategy to design the CCFA effectively provides the model with a global receptive field along the spatial direction and also reduces the number of parameters, which further improves the performance and efficiency of the model.

3) Cross-Layer Spatial–Frequency Attention: By performing frequency interactions among spatial tokens within groups, CSFA effectively leverages multiscale features' spatial–frequency information. This enables the model to precisely capture spatial localization cues of targets, thereby improving positioning accuracy. Based on the results of the fourth group of experiments, adding CSFA to the feature

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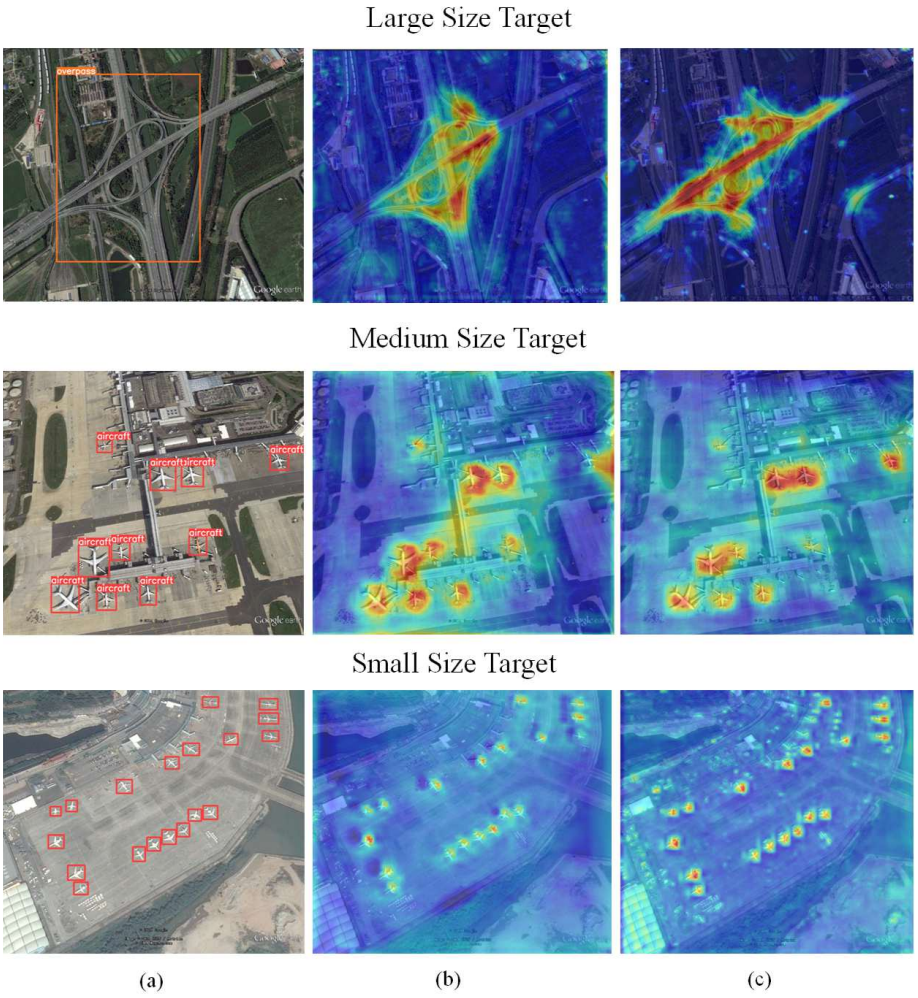


Fig. 11. Small, medium, and large target heat maps generated by baseline networks and SAFF-DETR. (a) Original image. Heat map generated by (b) baseline network and (c) SAFF-DETR.

fusion stage improves model accuracy from 92.1% to 92.5% while reducing model parameters by 7.9%, validating the effectiveness of CSFA. This improvement is because the CSFA provides the model with a global receptive field along the channel dimension, which enhances feature representation and optimizes parameter efficiency.

4) Effect of α: We investigate the impact of the high- and low-frequency interaction ratio α on model performance under the CCSFF framework, as shown in Fig. 9, where a larger α indicates more low-frequency interactions. As shown in the figure, we find that a larger α helps to reduce the model's parameter count, and the model's accuracy initially increases and then decreases as α increases, reaching its peak at $\alpha = 0.5$. When $\alpha = 0$, the model performance is poor, as in this case, only high-frequency interactions are performed. We speculate that low-frequency interactions play an important role in CCSFF. However, when only low-frequency interactions are performed at $\alpha = 1$, the performance difference compared with $\alpha = 0.5$ is about 0.3%, further indicating that high-frequency interactions are also indispensable in feature fusion. Overall, the effective synergy between high- and low-frequency interactions is key to improving model performance.

Furthermore, through systematic parameter search, we observed that the optimal frequency interaction ratio α decreases to 0.25 on small-target-dominated datasets, indicating the model's stronger reliance on high-frequency feature interaction. Building upon this finding, we speculate a significant correlation between target size distribution and the optimal frequency interaction ratio. Specifically, datasets dominated by small targets require enhanced high-frequency feature interaction, whereas those dominated by large targets benefit from strengthened low-frequency feature interaction. To ensure fairness and comparability in our comparative experiments, we consistently maintained a fixed frequency interaction ratio of $\alpha = 0.5$ throughout all subsequent experiments.

D. Comparison Analysis

In order to verify the effectiveness and robustness of the proposed algorithm, we conducted comparative experiments with other SOTA models on NWPU VHR-10, RSOD, and the small-target dataset VEDAI. It is noteworthy that in large-scale datasets, objects of the same category may exhibit significant

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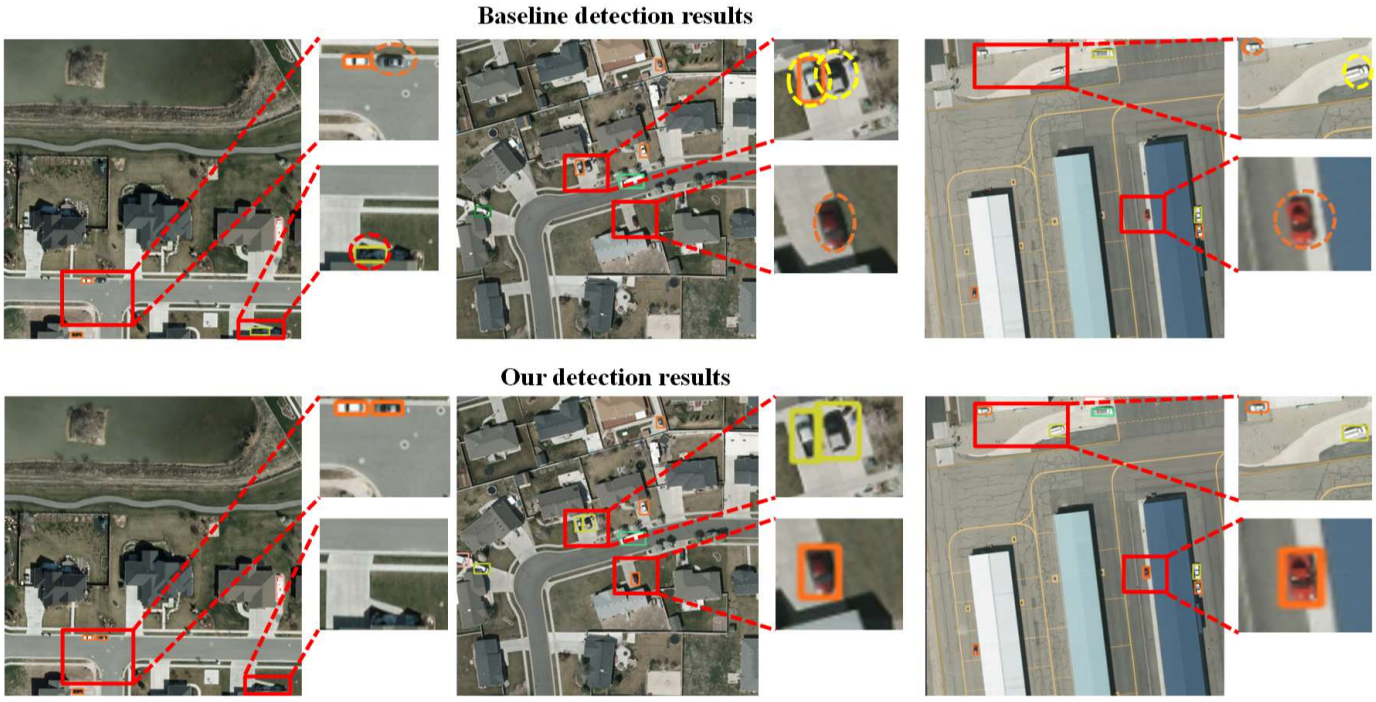


Fig. 12. Comparison of baseline and SAFF-DETR small object detection results on VEDAI. The top is the object detection result of the baseline network, and the bottom is the object detection result of the SAFF-DETR. The dashed circles within the red zoomed-in-view boxes indicate missed or false detections for the corresponding categories.

size variations, making it impossible to define multiscale targets based solely on category information. Therefore, we employ the universal mAP metric for overall performance evaluation, supplemented by instance-level pixel-area classification to calculate corresponding mAP scores, thereby comprehensively assessing the model's multiscale detection capability. For small-target datasets, all object instances inherently meet the size criteria for small targets (with an average area smaller than 20×20 pixels), meaning all categories can be uniformly treated as small targets.

1) Detection Results on NWPU VHR-10: As shown in Fig. 7, the NWPU VHR-10 dataset has a wide range of target size distribution, which can reflect the network's capability in multisize object detection. We use mAP_{50}, $mAP_{50:95}$, and the mAP value for each category as evaluation metrics, while also comparing the model's parameters and GFlops. The results are shown in Table III. On the testing set, SAFF-DETR achieves the best performance compared with several SOTA methods. Specifically, mAP_{50} and $mAP_{50:95}$ reach 93.3% and 61.5%, respectively. Compared with the baseline model, we successfully improved performance by 1.6% and 1.1%, with only a 0.7M increase in model parameters. Compared with the best single-stage algorithm, the performance is improved by 1.6% and 6.6%, respectively, and compared with the best end-to-end detector of the same type, the performance is improved by 1.3% and 2.5%. Although the proposed model exhibits moderately higher parameter size and computational complexity compared with specialized lightweight models, it achieves significant detection accuracy improvements. Compared with the best lightweight model, it achieves 3.1% higher mAP_{50} and 3.9% higher $mAP_{50:95}$. This design maintains a strong balance between accuracy and efficiency within acceptable computational costs, offering a better solution for high-performance remote sensing detection platforms.

Fig. 10 shows a comparison of detection results between the baseline model and our proposed model on the NWPU VHR-10 dataset. It can be observed that the baseline model misdetects the chimney in the first image as an AC and a CA, while our network avoids this issue. In the second and third images, the baseline network exhibits missed detections and false detections, respectively, whereas our network correctly identifies the targets, further demonstrating the superior performance of our method.

2) Detection Results on RSOD: As shown in Fig. 7, compared with the NWPU VHR-10 dataset, the RSOD dataset exhibits a wider range of object size variations. To more comprehensively evaluate the model's capability in detecting objects of different scales, we introduce refined evaluation metrics: mAP_S, mAP_M, and mAP_L, which correspond to small, medium, and large objects, respectively. These metrics impose higher demands on the model's ability to handle significant variations in object sizes. Table IV shows the performance comparison between our proposed method and the latest methods on the RSOD testing set, where the model parameters and GFlops values are consistent with the results in Table III. Our method achieves the best performance in detection accuracy, particularly on the more stringent evaluation metric $mAP_{50:95}$, with a score of 69.3%, outperforming the second-best method by 1.3%. In addition, it improves by 3%, 0.4%, and 1.5% on the mAP_S, mAP_M, and mAP_L metrics, respectively, compared with the second-best method. To further validate the network's capability of detection with diverse target sizes, as shown

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in Fig. 11, we use Grad-CAM to generate heatmaps for the baseline network and SAFF-DETR in detecting small, medium, and large targets. The results show that our proposed method exhibits stronger target focus capability and better interpretability in detecting targets of different sizes.

3) Detection Results on VEDAI: Since the Transformer-based detector is inherently advantageous for large object detection, to further validate the capability of the proposed model for small object detection, we specifically conducted comparative experiments on the small-target dataset VEDAI. The results are shown in Table V, where SAFF-DETR achieves the best performance across all metrics, with mAP_{50} and $mAP_{50:95}$ reaching 48.7% and 29.8%, respectively. It is worth noting that among the nine categories of small targets included in the VEDAI dataset, our method achieves the highest detection performance in five of them. Specifically, for the BT, TR, and VA categories, SAFF-DETR improves the mAP by 2.5%, 19.4%, and 26%, respectively, compared with the baseline algorithm. Compared with dedicated small object detectors, SAFF-DETR also demonstrates a clear advantage, achieving 0.6% and 2.8% higher scores in mAP_{50} and $mAP_{50:95}$, respectively, than the best-performing small object detector. In Fig. 12, we visualize and compare the detection results of the proposed algorithm with the baseline network. It can be seen that SAFF-DETR performs exceptionally well in remote sensing small-target detection, effectively reducing missed detections and false detections. For example, in the first image, the baseline algorithm fails to detect the small car on the roadside and misdetects the shadow near the house as a TR, while our algorithm not only correctly detects all targets but also avoids misjudging the shadow. In the remaining two images, the baseline network exhibits both missed and false detections, whereas SAFF-DETR accurately completes the target detection task, further validating its superior performance in small-target detection.

V. Conclusion

In this article, we propose an efficient end-to-end detector named SAFF-DETR for remote sensing images with a wide range of target sizes.

First, to address the inherent limitation of Transformer-based detectors, which exhibit strong perception for large targets but insufficient detection capability for small ones, we design the MRF module. Through its placement strategy and multibranch architecture, it effectively enhances the representation of small object features without significantly increasing the computational burden. Moreover, we introduce two carefully designed frequency-attention mechanisms, CCFA and CSFA, during the multiscale feature fusion stage. These mechanisms promote the network's exploitation of frequency information, enabling the capture of contextual information from multiple perspectives while preserving linear computational complexity.

Building on this, to overcome the limitations of traditional multiscale feature fusion methods, such as feature pyramids, which involve complex upsampling and downsampling operations leading to feature distortion and loss of varying-sized targets, as well as reliance on layer-by-layer information transmission mechanisms, we introduce the CCSFF structure. By combining CCFA and CSFA, CCSFF enables multiscale feature interaction without the need for upsampling and downsampling. It enables efficient cross-layer interaction in a single step, while significantly reducing computational costs. Benefiting from the overlapping window strategy of different scales and the global modeling capability of frequency-attention mechanisms in the CCSFF structure, the network's ability to detect targets with significant size variations is effectively improved.

Although SAFF-DETR achieves promising performance in multisize remote sensing detection tasks, it still has certain limitations.

1) In complex scenarios, a fixed interaction ratio α struggles to adapt to unknown target size distributions. Future research should incorporate either adaptive α modulation or learnable dynamic weighting mechanisms to automatically optimize the high–low-frequency interaction ratio, thereby enhancing model generalization capability.
2) The method has been validated on high-resolution remote sensing datasets, demonstrating its effectiveness. However, its performance on low-quality imaging data requires further investigation. Future work could leverage the complementary advantages of multisource data to overcome the limitations of single data sources in imaging quality and feature representation, thereby providing a viable solution to these challenges.

Thereby, in the future, building on this work, we will explore fusion detection methods for multisource remote sensing images to further improve the accuracy of remote sensing object detection. In addition, cross-modal collaborative detection will be a key focus of our research.

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